Tutorial - 04

Q) A University wants to predict the finale exams score of students based on their academic and behavioural attributes. Data from 10 students has been collected.

Input features :- Studyhours / week
Attendance %
Assignment score (out of 100)
No. of practice tests taken

Target variable:
Finale Exam Score (out of 100)

Studyhours	Attendance	Assignment Score	Practice tests	Final Exam Score
12	85	78	3	80
15	90	82	4	85
10	75	80	2	72
18	95	88	5	90
8	70	65	1	68

Q) 1) Identify the independent and dependent variables.

Independent variables (x)
→ study hours, attendance, assignment score, practice tests.

Dependent variables (y) Final Exam Score

2) Write multiple linear regression equation

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4$$

Final score = $B_0 + B_1$ (Study hours) + B_2 (Attendance) + B_3 (Assignment Score) + B_4 (practise tests)

2) Explain the impact of attendance on exam score

As attendance is increasing score is also getting increased, there is a positive relationship b/w attendance & exam score ($r_2 > 0$)

4) State one assumption of linear regression in this scenario find MSE for 5 iterations

assume is linear relationship b/w input features & final exam score

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y_i - \hat{y})^2$$

assume weights = 0.01

$$w_i = w_i + \alpha E (\text{Error} - x_i)$$

error = $y - \hat{y}$

$w_0 = w_0 - w_2 + w_3 w_1$
$= 0$

y	$\hat{y}$	error
80	0	80
85	0	85
72	0	72
90	0	90
68	0	68
		395

$\hat{y} = w_0 + w_1 x_1 + w_2 x_2 + \dots$
$0(12) + 0(15) + 0(10) + 0$

$\boxed{\hat{y} = 0}$

$$MSE = \frac{1}{n} \sum_{i=1}^{n} (y - \hat{y})^2$$

$w_1 = 0 + 0.01 (395(12) + 395(15) + 395(10) + 395(18) + 395(8))$

$= 0 + 0.01 (24885)$

$w_1 = 248.85$

$w_2 = 0 + 0.01 (395(85) + 395(90) + 395(75) + 395(95) + 395(90))$

$= 0.01 [163925]$

$= 1639.25$

$w_3 = 0 + 0.01 (395(78) + 395(82) + 395(70) + 395(88) + 395(65))$

$w_4 = 0 + 0.01\,(395(3) + 395(4) + 395(2) + 395(5)$
$+ 395(1)]$
$= 0.01 \times 5925$
$= +9.25$

$\hat{y} = 3.95 + 248.5\,x_1 + 1639.25\,x_2 + 1512.85\,x_3 +$
$+ 9.25\,x_4$

$\hat{y} = 3.95 + [(248.5)12 + (639.25)85 + (1512.85)(78)$
$+ (59.25)3]$

$\hat{y} = 260506.45$

$\hat{y} = 3.95 + 248.5(15) + 1639.25(90) + 1512.85(82) +$
$= 275554.65$

$\hat{y} = 3.95 + 248.5(10) + 1639.25(75) + 1512.85(70)$
$+ 59.25(2)$

$\hat{y} = 231450.7$

$\hat{y} = 3.95 + 248.5(18) + 1639.25(95) + 1512.85(88)$
$+ 59.25(5)$
$= 293692.75$

$\hat{y} = 3.95 + 248.5(8) + 1639.25(70) + 1512.85(65)$
$+ 59.25(1)$
$= 215433.95$

y	$\hat{y}$	$error(y-\hat{y})$	$(y-\hat{y})^2$
80	260506.45	-26042645	678219358 49.60
55	2755554.65	-2755,469.65	7592629920 71
72	231450.7	-231378.7	53536102813.69
90	293632.75	-293542.75	86167346077 56
68	215133.95	-215065.95	46253362849.40

$$MSE = \frac{1}{n} \Sigma (y-\hat{y})^2$$

$$MSE = \frac{1}{5}(7,846,391,739,671)$$

$$= 1,569,278,347,934$$